

PROJECT PROPOSAL

1. Project Title

Medical Report Analyzer

An AI-powered web application that transforms complex medical reports into clear, plain-language summaries accessible to everyone.

2. Problem Statement

Medical reports are filled with clinical terminology, abbreviations, and technical jargon that is difficult for the average person to understand. Patients receiving blood test results, discharge summaries, lab reports, or prescriptions are often left confused about what the findings mean and what actions they should take.

Why the problem matters:

- Millions of patients receive medical reports each year without the ability to understand them fully.
- Misunderstanding a report can lead to delayed treatment, unnecessary anxiety, or poor health decisions.
- Patients often rely on costly follow-up appointments just to have a doctor explain what the report says.

Why existing solutions are insufficient:

- Generic search engines return broad medical information, not personalized analysis of a specific report.
- Rule-based tools cannot handle the variability in report formats, abbreviations, and layouts across hospitals and labs.
- No current mainstream tool offers a conversational, interactive way to ask questions about your own report.

Why AI is relevant:

- Large language models can read unstructured medical text, understand clinical context, and rephrase findings in plain language.
- Conversational AI enables a natural Q&A experience that static tools simply cannot replicate.

3. Objectives

The primary goals of the Medical Report Analyzer are:

- Build a user-friendly web application where patients can input or upload any medical report.
- Use AI to automatically generate a plain-language summary of the report content.
- Extract and highlight key findings such as diagnoses, medications, lab values, and abnormal results.
- Provide an interactive chatbot so users can ask specific questions about their report.
- Securely store all reports, summaries, and chat history per user account for future reference.
- Support both text-based and image/scanned reports using OpenAI's vision capability.
- Ensure secure, per-user access through a robust authentication system.

4. Proposed Features

Authentication System

- User registration and login with JWT-based secure sessions.
- Private, per-user data isolation ensuring reports are only accessible by their owner.

Report Input & Upload

- Text input field for pasting report content directly.
- File upload support for PDF and image-based reports.
- OCR and vision processing for scanned documents via OpenAI Vision.

AI-Powered Analysis Dashboard

- Automatically generated plain-language summary of the report.
- Structured extraction of key findings: diagnoses, lab values, medications.
- Flagging of abnormal results for immediate user attention.

Interactive Q&A Chatbot

- Built-in chatbot allowing users to ask follow-up questions about their report.

- Context-aware responses grounded in the specific report content.
- Conversational interface for a natural user experience.

Report History

- All past reports, AI summaries, and chat conversations saved per account.
- Users can revisit any prior report and continue asking questions.

5. Technology Stack

Layer	Technology	Purpose
Frontend	React.js	Upload page, summary dashboard, chatbot panel, report history, auth
Deployment (Frontend)	Vercel	Host the React app
Backend	Node.js + Express.js	REST API: /upload, /summarize, /chat, /reports, /auth
Deployment (Backend)	Render	Host the Express server
Database	PostgreSQL via Supabase	Tables: users, reports, summaries, chat_history
File Storage	Supabase Storage	Store uploaded PDF/image files
AI	OpenAI	Summarization, Q&A chatbot, key entity extraction, vision/OCR
Auth	JWT (JSON Web Tokens)	Secure per-user sessions

6. Expected Outcome

Upon completion, the Medical Report Analyzer will provide the following tangible benefits:

For Users:

- Any patient, regardless of medical background, will be able to understand their own health reports instantly.
- Users will receive a structured, readable summary within seconds of submitting their report.
- The chatbot will answer personalized questions, reducing the need for follow-up consultations for basic clarification.

- Secure report storage means users have a personal health record accessible at any time.

Workflow Improvements:

- Eliminates the need to search generic health websites for medical term definitions.
- Reduces anxiety caused by unread or misunderstood reports by making findings immediately accessible.
- Supports both typed text and uploaded images, covering all common report formats.

Benefits of AI Integration:

- OpenAI handles the full complexity of medical language without hardcoded rules or templates.
- Vision AI removes the limitation of text-only tools, processing scanned and photographed documents seamlessly.
- The conversational chatbot creates a personalized experience that static report tools cannot offer.
- AI-based extraction ensures accuracy across the wide variety of formats used by different hospitals and laboratories.